When mutualism turns costly: abiotic stress shifts endophyte benefits in cool-season grasses

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[1]*If the data are not on github as csv's then I would create a temporary dropbox link so that reviewers could reproduce the analysis.*

Abstract

Benefits of plant-microbe symbioses are predicted to increase under environmental stress, but the relative roles of abiotic and biotic factors as modifiers of symbiotic interaction outcomes remain unclear. We studied how the independent and combined effects of aridity and herbivory modify the demographic effects of seed-transmitted fungal endophytes on three native grass host species using geographically distributed field experiments along a precipitation gradient, crossed with herbivore exclusion. Host fitness declined under increasing abiotic stress, and endophyte-symbiotic grasses often fared worse than non-symbiotic ones[2], suggesting that interactions that are beneficial under benign conditions can become costly under stress. Herbivory had no detectable effect on host demography or endophyte-mediated outcomes.[3] Overall, abiotic stress was the primary driver of mutualism outcomes between plants and fungal symbionts, but in the opposite direction as predicted by theory and prior work. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

[2] *I know space is tight but can we make this a more specific or quantitative summary of the results?*
[3] *I think this is not true in the new results...?*

Introduction

Symbioses are widespread and ecologically important, occurring across both plants and animals, and often mediating key ecological functions such as stress tolerance and defense against natural enemies. These interactions are famously context-dependent, with the direction and strength of outcomes shaped by the environment in which they occur (Bronstein, 1994; Hoang et al., 2024; Hoeksema and Bruna, 2015). According to the stress-gradient hypothesis (SGH), positive interactions are generally expected to increase under stressful conditions (Bertness and Callaway, 1994; Holmgren and Scheffer, 2010; Louthan et al., 2015), but actual outcomes depend on the type and intensity of biotic and abiotic stress. For example, microbial symbionts that benefit host plants under drought stress may become neutral or even parasitic under more benign conditions (Giauque et al., 2019).

Because microbial symbioses are expected to become more beneficial in stressful conditions, they can substantially enhance host resilience to environmental stress. Under biotic stress, such as herbivory, microbial symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing herbivore growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly, under abiotic stress, such as low precipitation, nutrient limitation, or high temperature, symbiotic microbes can enhance host tolerance through a variety of mechanisms (Afkhami and Rudgers, 2008; Clay and Schardl, 2002). Symbionts including mycorrhizal fungi, rhizobia, and endophytic bacteria can improve water and nutrient acquisition by increasing root surface area, facilitating nutrient mobilization, or promoting nitrogen fixation, allowing plants to maintain growth and metabolic function under drought or poor soil conditions (Amijee et al., 1989). In addition, many symbiotic microbes stimulate the production of antioxidant enzymes, which reduce oxidative damage caused by stressors like drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023).

While stress is hypothesized to strengthen the benefits of symbiotic microbes, growing evidence suggests that it could also elevate costs of symbiosis. Maintaining symbionts requires hosts to allocate valuable carbon and nutrients to sustain their microbial partners. For instance, mycorrhizal fungi may require up to 20% of photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly, vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen et al., 2004). For example, under low nutrient or water conditions, endophyte-infected grasses may produce fewer tillers and lower total biomass compared to endophyte-free plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002)

Despite growing interest in the role of symbiosis in shaping host demography, few studies have tested how abiotic and biotic stressors synergistically influence the demographic effects of symbionts along environmental gradients. Most work has been conducted in controlled or common garden experiments. In *Ammophila breviligulata*, herbivory but not drought reduced growth, and no interaction was observed; however, plants hosting endophytic fungi were more resistant to these stressors (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function. Understanding how abiotic stress, biotic stress, and microbial symbiosis interact to influence plant performance requires experiments that jointly vary all three factors. However, to our knowledge, no such comprehensive study has been conducted. Here, we address this gap by examining the effects of biotic and abiotic stress on grass-endophyte symbioses in three grass species spanning a natural range-core/range-edge precipitation gradient.

Fungal endophytes of plants provide a tractable system for studying the ecological effects and context-dependence of host-symbiont interactions, as host status (symbiotic vs symbiont-free) can be experimentally manipulated. A well-studied example is the association between cool-season grasses and seed-transmitted fungal endophytes in the genus *Epichloë* (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Because these fungi are vertically transmitted, the fitness of host plants is closely linked to that of their symbionts, which is expected to favor the evolution of mutualistic interactions (Gundel et al., 2011, 2024). In many cases, endophyte-symbiotic grasses outperform endophyte-free grasses under water limitation, although the benefits depend on host and endophyte identity (Decunta et al., 2021). Beyond these performance effects, infection frequency itself is shaped by environmental conditions, including long-term variation in precipitation and temperature (Fowler et al., 2025; Rudgers et al., 2016) Even when endophytes do not enhance survival under stress, they may still boost host fitness by promoting growth and reproduction (Yule et al., 2013). More generally, prior work has shown that effects of endophyte symbiosis may vary in magnitude and even sign across host vital rates (Rudgers et al., 2012), which argues for considering multiple life history pathways. Importantly, hosting endophytes can entail costs, including the allocation of plant carbon to sustain the fungal partner and potential risks associated with horizontal transmission. These costs may be offset by benefits such as herbivore deterrence, mediated by fungal alkaloids that protect against both invertebrate and vertebrate herbivores (Clay et al., 2005; Saikkonen et al., 2010) Together, these features make grass–*Epichloë* symbioses an ideal system for investigating how abiotic stressors, such as drought, interact with biotic stressors, like herbivory, to determine the outcome of symbiosis.

Across an aridity gradient in the south-central United States, we investigated how endophyte symbiosis affects multiple host vital rates under varying abiotic and biotic stress. This gradient, from mesic conditions in the east to increasingly arid conditions in the west, provides a biologically realistic stress gradient, and coincides with the westernmost range limits of many eastern North American grass species. At each of seven sites along the gradient, we manipulated vertebrate and invertebrate herbivory (allowing herbivore access versus excluding herbivores) to test its interaction with abiotic stress. We also examined the effects of Epichloë symbiosis by comparing endophyte-symbiotic and endophyte-free individuals across three species of native cool-season grasses.

We addressed how variation in precipitation and herbivory influences the outcome of grass–*Epichloë* symbioses. Specifically, we asked: (i) under reduced precipitation, do endophytes enhance host fitness by mitigating drought stress (stress-amelioration scenario), or can they impose a net cost under extreme combined stress from drought and herbivory (reduced-benefit scenario)? and (ii) does the relative importance of endophyte-mediated benefits change along the stress gradient, as predicted by the stress-gradient hypothesis (SGH), with higher abiotic stress amplifying mutualistic effects? We predicted that endophytes would enhance host fitness under moderate drought in the absence of herbivores, representing an endophyte-driven stress amelioration scenario (Fig. 1A). When herbivores are added as a second stress under more extreme drought, the metabolic cost of maintaining the endophyte may reduce benefits or even impose a net cost, corresponding to a reduced endophyte-mediated benefits scenario (Fig. 1B). Overall, consistent with the SGH, endophyte-mediated benefits are expected to increase under lower precipitation, with the presence or absence of herbivores modulating these effects.

Materials and methods

Study system

Our study focused on three cool-season perennial grasses native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (Poaceae, subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a small, short-lived perennial that germinates from autumn to late winter and typically flowers between March and July. *E. virginicus* is a larger, long-lived species that begins flowering as early as March or April and continues throughout the summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers from late spring through summer. *Agrostis hyemalis* hosts the endophyte *Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* hosts *Epichloë PauTG-1* or *E. typhina subsp. Poae* (Gundel et al., 2020). Like other *Epichloë* species, these endophytes are primarily transmitted vertically through seeds and are additionally capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). Herbivory in this system is primarily from generalist insects and small mammalian grazers, which can influence both plant performance and endophyte-mediated defenses.

Source populations and symbiont removal

Plants used in our experiment were derived from natural (source) populations of each species (Fig. 2, Table S1). Prevalence of fungal endophytes in these natural populations ranged from XX% to XX% endophyte-positive hosts (NEW SUPP TABLE). Seeds from these source populations (from ca. 30 plants per population) were either heat-treated or left untreated, intended to produce symbiont-free (E^-) and symbiotic (E^+) plants from the same genetic lineages. For *Elymus virginicus* and *Poa autumnalis*, heat-treated seeds were placed in a drying over at 60°C for approximately five days. For *Agrostis hyemalis*, seeds were heat-treated in a water bath at 60°C for 12 minutes. [4] All seeds, both heat-treated and non-heat-treated, were surface-sterilized with 10% bleach solution then germinated on agar plates. Seedlings were transferred to potting soil and grown in the Rice University greenhouse. Once these plants were sufficiently large in size they were vegetatively propagated, with clonal identities tracked across individuals. [5] Before transfer to the field experiment, we confirmed endophyte status (E^+ or E^-) of all seedlings using either microscopy or an immunoblot assay. [6] For microscopy, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae [7]. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to

[4] *We should talk about this. You need more detail here – I have tried adding some but there is more to do. They were all treated but not all treatments were successful, which we need to report. And there were a few different methods used on different species. This should all go in the supplement. It is also not true that heat treatment has no effect on seed viability, which is why we tried to use seeds that were multiple generations removed from the heat treatment.*

[5] *Add a sentence giving basic descriptive statistics of how many genotypes and clones were used for each species.*

[6] *Realized numbers of E+ and E- per species per population should be provided in a suppl table.*

[7] *cite*

visually indicate presence or absence (**?**). Both methods yield similar detection rates.[8] For genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assume that its status applies equally to all clones of a given source genotype. Greenhouse-raised plants were planted into the field experiment in February-March 2023.

Common garden experimental design and climate data

To examine the effects of endophyte symbiosis across an abiotic stress gradient, we established common garden experiments with E+ and E- hosts at seven sites along an aridity gradient from Louisiana to central Texas, US (Fig. 2, Table S2).[9] From west to east, mean annual precipitation increases more than three-fold across these sites, while mean annual temperature is relatively consistent (Fig. S1).[10] For this reason, we focus on precipitation as the primary abiotic factor that varied across sites. This geographic gradient overlaps and exceeds the natural range edges of our focal species, ensuring realistic exposure to climatic and biotic conditions (Fig. 2A,B, C).

To characterize climate at these sites during the study period (2023-2025), we collected daily temperature and precipitation data from PRISM (Hart and Bell, 2015). These daily values were used to calculate the mean temperature (°C) and cumulative precipitation (mm) for each year, spanning from the time plants were transplanted to when demographic data were collected.[11] During our study period, realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites (Fig. 2D).

At each site along the gradient, we established 24 square plots (1.5 m $\times$ 1.5 m), eight per host species. Within each site, each plot was randomly assigned to either herbivore access or herbivore exclusion (Fig. 2E). The herbivory treatment targeted both vertebrate and invertebrate hebivores: exclusion plots had fencing on four sides and were sprayed with Sevin insecticide (active ingredient Zeta-Cypermethrin) 2-3 times per growing season following the manufacturer's protocol, and access plots were fenced on two sides to control for any fence effect. To ensure that our herbivory treatment reflected the level of herbivory experienced by individual plants, we counted the number of damaged plants per plot and estimated the

[8] *What is the evidence for this statement?*

[9] *Figure comments: can you spell out 'precipitation' rather than 'P'? Change gray dots to 'GBIF occurence'. Change 'Garden' to 'Experimental site'. In D, I would order sites from west to east, rather than precip rank. Legend needs to describe panels E-F. Unclear what the green and red colors in E are meant to show - this may get confused for E+/E-. I like the herbivory figure – it would be good to see this broken up by species, maybe as a supp figure.*

[10] *Unclear what you are showing here. Are these 30-year normals or realized weather? You say annual mean precip but this looks exactly like Figure 2D.*

[11] *I am a little confused here because the figure says this is cumulative precipitation over the study period, but here you say you calculated precip per year, so is this an average over years? And when you say 'year', is this the calendar year or the census year?*

proportion of damaged plants per site and herbivore treatment. Our results confirmed that the herbivory treatment was effective (Fig. 2F).[12]

Each plot was planted with 15 individuals, a mix of E+ and E- host-plants. Initial endophyte prevalence varied from 20% (3 E+, 12 E-) to 80% (12 E+, 3 E-); this aspect of the experimental design was intended to address a different question about change in symbiont prevalence across generations, and is not further considered here. For each species at each site along the precipitation gradient, we planted a total of 60 E+ and 60 E- host-plants, half of those experiencing herbivore access and half experiencing herbivore exclusion.

[13]

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual, survival was recorded as alive or dead and, for surviving plants, size was measured as the count of tillers. Growth was quantified as the log ratio of tiller count between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in years t and $t+1$. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*. In *A. hyemalis*, one spikelet yields one seed, while in *P. autumnalis* and *E. virginicus* a single spikelet can produce 2-6 seeds.

Statistical modeling

To assess whether endophytes confer benefits or impose costs under abiotic and biotic stress, we fit generalized linear mixed models for each vital rate response in a hierarchical Bayesian statistical framework. All models shared the same hierarchical structure for their expected values and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, inflorescence count with a zero-inflated negative binomial distribution, and spikelet count with a negative binomial distribution. All models included species-specific fixed effects for symbiosis (endophyte presence/absence), precipitation, and herbivory (herbivore access/exlusion), including all relevant two and three-

[12] *I recommend moving the methods to data collection methods below (you will also need to define what you mean by "damage" and move the result to results).*

[13] *Kerrville and Sonora sites were destroyed and replanted in 2024. That should be described here. Also, POAU was not planted in Sonora. The map is accurate but you need to describe that here.*

way interactions. We considered models with linear and quadratic forms for the influence of precipitation, to account for the possibility of non-monotonic responses. We present the quadratic model here (Eq. 1) because it encompasses the linear model as a special case.

For each vital rate model, the expected value for the i^{th} observation (μ_i) was given by:

$$\begin{aligned}
\mu_i = \beta_0[Sp_i] + \beta_P[Sp_i]precip_i + \beta_{P^2}[Sp_i]precip_i^2 + \beta_E[Sp_i]endo_i + \beta_H[Sp_i]herb_i \\
+ \beta_{EP}[Sp_i]endo_i precip_i + \beta_{HP}[Sp_i]herb_i precip_i \\
+ \beta_{HE}[Sp_i]herb_i endo_i + \beta_{HEP}[Sp_i]endo_i herb_i precip_i \\
+ \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i],Sp_i} + \rho_{\text{plot}[i],Sp_i} \\
\phi_{\text{site}[i],Sp_i} \sim N(0, \sigma^2_{site}) \\
\rho_{\text{plot}[i],Sp_i} \sim N(0, \sigma^2_{plot}) \\
\omega_{\text{source}[i],Sp_i} \sim N(0, \sigma^2_{source})
\end{aligned} \tag{1}$$

Here, $precip_i$ is total precipitation (mm)[14] accumulated over the 12 months preceding the census, $endo_i$ indicates endophyte presence ($endo_i = 1$) or absence ($endo_i = 0$), and $herb_i$ indicates herbivore access ($herb_i = 1$) or exclusion ($herb_i = 0$)[15]. The model includes random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity that is not explained by precipitation, $\rho_{\text{plot}[i],Sp_i}$ captures plot-to-plot variation within sites, and $\omega_{\text{source}[i],Sp_i}$ captures variation associated with the provenance of transplants. Random effect for site, plot, and source are host species-specific but the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species; this assumes, for example, that site-to-site heterogeneity is of a similar magnitude across species, even as the model allows for "better" and "worse" sites to differ by species. This hierarchical structure allows the model to"borrow strength" across species, improving estimates of fixed effects while still capturing species-specific responses to environmental gradient.

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024).[16] We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using a posterior predictive check (Fig. S2).

Next, for each vital rate, we compared candidate models with a linear or quadratic influence of precipitation using leave-one-out cross-validation (LOOCV) to identify the best-supported model (Vehtari et al., 2017). LOOCV combines validation and training by

[14] *I think we should update FIgure 2 so that it shows precipitation of both years, since the combined effects of site and year is what gives us greater ppt coverage. Also, was ppt scaled to mean zero for analysis?*

[15] *Is this right? We need to make sure we interest the fence effects correctly.*

[16] *Need to say more about priors, chains, thinning, etc.*

iteratively holding out one observation as a test set while fitting the model on the remaining $n-1$ observations. This process is repeated n times, once for each observation, resulting in n fitted models (Silva and Zanella, 2024). The overall test error estimate is then obtained by averaging the prediction errors across all n models (Eq. 2) for each vital rate:[17]

$$CV_n = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

Although we evaluated quadratic models, LOOCV showed that they provided minimal improvement over monotonic models (Table S3). Therefore, all subsequent analyses used the monotonic model, which captures the observed patterns while maintaining model simplicity.[18]

Drivers of endophyte effects on host demography

We used posterior parameter estimates to visualize predictions for how effects of symbiosis varied across abiotic and biotic contexts. For each vital rate, precipitation was summarized at key quantiles (10th, 25th, 50th, 75th, and 90th percentiles) to represent the range of site-level abiotic stress. [19] For each herbivory treatment (Access or Exclusion) at each level of precipitation, we computed posterior predictions of survival probability E^+ and E^- hosts using the best Bayesian model outputs (Eq. 1). [20] The difference ($\Delta = E^+ - E^-$) was calculated for each posterior sample to estimate median effects and 90% credible intervals. We also computed the posterior probability that $\Delta > 0$, representing the likelihood that endophytes confer benefits on vital rates under the given precipitation and herbivory conditions.

To quantify statistical support for abiotic and/or biotic contexts as modifiers of grass-enodphyte symbiosis, we fitted three hierarchical Bayesian candidate models for each vital rate (survival, growth, inflorescence, and spikelet production; Supplementary Methods S.1.1). All models included endophyte status as a baseline predictor but differed in whether they incorporated interactions with precipitation (Endophyte × Climate), herbivory (Endophyte × Herbivory), or both factors, including the three-way interaction (Endo-

[17] *In Eq 2, should CV_n instead be $LOOCV_n$? Also, you have not defined the y's in this equation, and I think it might be simpler to describe the method but not show the equation, since this is a standard model selection method and is not unique to our study.*

[18] *I think this is very rigorous. But it will be simpler for readers if we say from the beginning, where you first introduce the linear-quadratic contrast, that we tested both and found little support for quadratic effects (and reference supplement table), and then we can just show the linear model.*

[19] *I do not follow the significance or value of considering quantiles. Why not treat precipitation continuously? We an indicate quantiles on the x axis of a figure, if your idea is to represent the realistic range of ppt variation. Note that species would differ in their quantiles because only ELVI is at Sonora.*

[20] *I don't know what you mean by "best".*

phyte × Climate × Herbivory).[21] Random effects for site-year[22] ($\phi_{\text{site}[i],\text{Sp}_i}$), source population ($\omega_{\text{source}[i],\text{Sp}_i}$), and plot ($\rho_{\text{plot}[i],\text{Sp}_i}$) captured hierarchical variation while accommodating species-specific responses, as above. We compared candidate models using leave-one-out cross-validation (LOO-CV) (Vehtari et al., 2017), which estimates predictive accuracy by iteratively holding out each observation. Differences in expected log predictive density (ΔELPD), along with associated standard errors, were used to rank model performance.

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Wetter conditions promoted grass host survival for all species, and abiotic stress modified endophyte effects on survival for two of three host species. Grass-*Epichloë* symbiosis had mostly negative or neutral effects on survival at low precipitation, but increasingly enhanced survival under wetter conditions (Fig. 3; Fig. S3). tomCould you change the figures so that they say 'herbivore access' and 'herbivore exclusion' rather than fenced/unfences? Also, the legend says points are jittered but I don't think they are. The points get a little cutt off near the margins - could you add some space between panels? In *Agrostis hyemalis* and *Elymus virginicus* populations exposed to herbivores, median $\Delta(E^+ - E^-)$ ranged from approximately -0.026 to 0.019 under low precipitation ($\Pr(\Delta > 0) = 0.37$–0.61) and increased to 0.022–0.106 under high precipitation ($\Pr(\Delta > 0) = 0.70$–0.85; Table S4). In fenced plots, positive effects appeared only at the highest precipitation for *Agrostis* (median $\Delta = 0.011$–0.038, $\Pr(\Delta > 0) = 0.59$–0.72), while for *Elymus*, strong positive effects were evident at high precipitation (median $\Delta = 0.089$–0.113, $\Pr(\Delta > 0) = 0.91$–0.95). *Poa autumnalis* showed little survival benefit in unfenced plots across the gradient (median $\Delta = -0.056$ to -0.002, $\Pr(\Delta > 0) = 0.14$–0.43), but weakly positive effects emerged when herbivores were excluded (median $\Delta = 0.002$–0.026, $\Pr(\Delta > 0) = 0.57$–0.71). [23]

Epichloë endophyte presence generally enhanced relative growth of individuals across the precipitation gradient, with benefits increasing under wetter conditions and not modulated by herbivory (Table S5).[24] In *A. hyemalis*, effects were strongly positive even at moderate precipitation in unfenced plots (median $\Delta = 0.370$–0.385; $\Pr(\Delta > 0) = 0.946$–0.992), and

[21] *This is the same as Eq 1, yes?*

[22] *Here you say site-year and above you say site. WHich is it?*

[23] *I am working quickly through the results so that I can get this back to you, but I think we need to be a little more careful in describing costs and benefits since the credible intervals for delta mostly overlap zero. I can help with this on the next edit.*

[24] *This is not what I see in the figure. For AGHY and ELVI, it looks like benefits were stronger under herbivore access (the direction I would have expected) and not affected by precipitation.*

similar trends were observed in fenced plots though slightly weaker (median $\Delta = 0.080$–0.262; $\Pr(\Delta > 0) = 0.661$–0.878). *E. virginicus* showed small positive or neutral effects under dry conditions (median $\Delta = 0.232$–0.259; $\Pr(\Delta > 0) = 0.869$–0.953), which became more pronounced at higher precipitation (median $\Delta = 0.124$–0.134; $\Pr(\Delta > 0) = 0.779$–0.790). In *P. autumnalis*, growth was context-dependent: unfenced plots shifted from negative to positive effects across the precipitation gradient (median $\Delta = -0.275$ to 0.144; $\Pr(\Delta > 0) = 0.169$–0.775), while fencing generally amplified benefits at low to moderate precipitation (median $\Delta = 0.020$–0.390; $\Pr(\Delta > 0) = 0.559$–0.927), but became negative at higher precipitation (median $\Delta = -0.262$ to -0.322; $\Pr(\Delta > 0) = 0.056$–0.072).[25]

Epichloë endophytes consistently increased inflorescence production across the precipitation gradient (Fig. 5).[26] By contrast, spikelet production was negatively affected by *Epichloë* under low precipitation but increased under wetter conditions, with effects varying among species and depending on herbivore exclusion (Fig. S6; Fig. S7). Inflorescence benefits in *A. hyemalis* and *E. virginicus* were minimal under dry conditions but increased with precipitation, with only minor influence from herbivore exclusion. [27] In *P. autumnalis*, strong inflorescence benefits occurred under fencing at moderate-to-high precipitation (median $\Delta = 0.47$–1.54; $\Pr(\Delta > 0) = 0.96$–1.00; Table S6). Spikelet production showed a similar pattern: *E. virginicus* and *P. autumnalis* experienced weak or negative effects under low precipitation (median $\Delta = 0.10$–0.40; $\Pr(\Delta > 0) = 0.23$–0.40), shifting to positive effects at higher precipitation, with fencing amplifying these benefits across the gradient (median $\Delta = 0.95$–1.56; $\Pr(\Delta > 0) = 0.79$–0.99; Table S7).

Across fitness components, *Epichloë* endophytes generally enhanced host performance under wetter conditions, whereas their effects were costly or occasionally neutral under drier conditions, consistent with the inhibition scenario (Fig. 1). Endophyte-mediated benefits were consistent regardless of herbivore access. [28]

Abiotic factors as the primary drivers of fitness components across the stress gradient

The outcomes of grass-endophyte symbioses – whether beneficial, costly, or neutral – was primarily modulated by climatic variation rather than by herbivore pressure. Across all vital

[25] *I would be careful here because you have extrapolated the Delta functions beyond the observed ppt range.*

[26] *The figure suggests this was only true for POAU.*

[27] *I don't think we can have much confidence that ELVI experienced stronger benefits under wetter conditions. Also, I would free the scales on the y-axes since different species produce different numbers of infs as a basline.*

[28] *I moved this sentence for now. This is probably not the best place for it, and I wonder if we should use a sentence like this to lead off the discussion, rather than using it in the results section. My other concern with this sentence, and with parts of the results, is that this sounds really straightforward, whereas I think the results are actually more complex than this.*

rates (survival, growth, inflorescence, and spikelet production), models comparing different drivers of the endophyte–host interaction consistently identified the *Endophyte* × *Climate* model as the best predictor of host demography (Table 1).[29] For instance, for survival, the *Endophyte* × *Climate* model outperformed the *Endophyte* × *Herbivory* model (ΔELPD = −2.1, SE = 2.8), and including the three-way *Endophyte* × *Climate* × *Herbivory* interaction further decreased predictive performance (ΔELPD = −5.0, SE = 1.9). Similar patterns were observed for growth, inflorescence, and spikelet production, indicating that climatic context is the primary driver of variation in the demographic consequences of endophyte symbiosis, whereas herbivory has weaker or context-dependent effects.

Discussion

Drawing on three years of field experiments and using Bayesian statistical models, we evaluated how biotic and abiotic stressors jointly shape the demographic effects of vertically transmitted *Epichloë* endophytes in three cool-season perennial grasses across a natural climatic gradient in the United States. Our results indicate that grass–endophyte symbioses generally enhance host performance under favorable (e.g. wetter) conditions, whereas their effects can be neutral or even detrimental under high abiotic stress, consistent with the inhibition scenario. Across species and fitness components, climatic variation emerged as the dominant driver of endophyte-mediated effects, while herbivory had a secondary or context-dependent influence. These results underscore that the outcome of grass–*Epichloë* symbiosis shifts along abiotic stress gradients (Bronstein, 1994; Saikkonen et al., 1998; **?**), providing empirical support for environmentally contingent mutualism in a broadly distributed host–symbiont system. [30]

Endophyte presence often enhances host performance under favorable or moderately stressful[31] conditions, a pattern widely documented across grass–*Epichloë* systems (**?**). Endophytes can increase host growth, reproduction, and resistance to herbivores (**?**), but these benefits are most apparent when the host has sufficient resources, such as water and nutrients, to support the fungus (**??**). Mechanistically, the host must allocate carbon and nutrients to maintain the endophyte, so under resource-rich conditions, the costs of symbiosis are outweighed by the benefits, resulting in a net positive effect. In our field experiment, we observed higher growth and survival of endophyte-infected grasses in wetter sites, consistent

[29] *I think the Methods section needs to provide more guidance on how readers should interpret ELPD and its SE. As I read it, these model rankings seem quite close.*

[30] *This paragraph is good but I think it needs to further emphasize that the influence of abiotic conditions is the opposite of what we predicted, and of what is expected under the SGH.*

[31] *unsure what point you are trying to make by combining favorable and 'moderately stressful' here. Also, I think it is important in the Discussion to distinguish between biotic and abiotic stress, since that is a theme of the paper.*

with previously documented positive effects of endophytes under favorable conditions (Afkhami et al., 2014; Rudgers et al., 2020).[32]

While endophyte infection enhanced host performance under favorable or moderately stressful conditions[33], its effects diminished and ultimately became costly under high abiotic stress, even in fenced plots where herbivores were excluded.[34] This pattern indicates that drought-driven resource limitation can impose costs on the symbiosis independent of biotic stress (Bronstein, 2001; Johnson et al., 1997; **?**). By contrast, other studies show that the context-dependent benefits of endophytes are often mediated by herbivore pressure (Bastias et al., 2017; **?**). For instance, in *Bromus setifolia*, endophyte infection is favored under high herbivore pressure, when defensive benefits outweigh the costs, but is selected against when herbivory is low or other costs are elevated (**?**). Our results suggest that similar context dependence can arise from abiotic factors, such as drought, independent of herbivory. Under low-precipitation conditions, reduced water availability constrains nutrient uptake and increases the relative metabolic burden of maintaining the endophyte, potentially tipping the cost–benefit balance against endophyte-symbiotic plants (Ahlholm et al., 2002; Bronstein, 2001; **?**). In addition, severe environmental stress can activate plant defense pathways-such as elevated phytohormone production and reactive oxygen species (ROS) that may inhibit endophyte growth or reduce its functional contributions to the host (**??**). Consequently, even in the absence of herbivores, endophyte-mediated benefits may become negative in resource-limited environments, consistent with the reduced performance of infected plants at the driest sites in our study. Although we found no evidence of symbiosis-related costs for inflorescences across the precipitation gradient, spikelet production showed a small but consistent cost at the driest sites. [35] This indicates that drought-driven resource limitation disproportionately affects certain reproductive components in infected plants, with different traits varying in their sensitivity to shifts in host–endophyte interactions (**?**).[36]

Across species and fitness components, climatic variation was the primary driver of endophyte-mediated effects, whereas herbivory played a secondary or context-dependent role. [37] Precipitation strongly shaped host demography in our system, with drier sites constraining

[32] *I am a little confused here because I thought the Intro was meant to set up the idea that stress can strengthen benefits of the symbiosis.*

[33] *Unclear what you mean by this.*

[34] *what's the relevance of this emphasis?*

[35] *Unclear what result this refers to.*

[36] *This also relates to the idea of demographic compensation, ie different vital rates responding differently to abiotic drivers - this may be worth bringing up in the Intro as a motivation for considering multiple vital rates side by side.*

[37] *Invoking context dependence here is a little "meta" since herbivory is itself context. So you are saying that the context dependence is context dependent! Also, I think we need to address the difference between fencing and herbivory. The herbivore effects was not consistent across sites because herbivore pressure varied, so the fence effect is dependent on baseline herbivory.*

survival, growth and fertility. Contrary to the predictions of the stress gradient hypothesis (SGH), which suggests that biotic interactions should become more positive under stressful conditions, endophyte-mediated benefits were reduced under high abiotic stress (Louthan et al., 2015; **?**). [38] Although most empirical tests of the SGH have focused on plant–plant interactions and have been developed and tested mainly by community ecologists (Bertness and Callaway, 1994), recent syntheses show that support for the SGH is actually strongest among microbial interactions (**??**), with bacterial systems[39] showing greater conformity to SGH predictions than any other kingdom[40] (**?**). This contrast highlights that plant–microbe mutualisms, including fungal endophytes, may not follow the same stress-dependency patterns documented in plant community ecology, and that endophyte benefits can decline rather than intensify under severe environmental stress.[41] A possible explanation for this opposite pattern is that the benefits of fungal endophytes are closely tied to host resource availability. Under high abiotic stress, host plants have limited resources for survival, growth, and reproduction, which can reduce the capacity of endophytes to confer benefits. Unlike plant–plant interactions, where neighbors can alleviate stress through mechanisms like shading or nutrient modification, endophytes depend on the host plant's internal physiological condition, such as its nutrient and water status, energy reserves, and overall health. When host resources are constrained, the costs of maintaining the symbiosis may outweigh its benefits, leading to reduced endophyte-mediated effects under high abiotic stress.[42]

[38] *This seems to me like the most important result that I expected would be more central to the first paragraph of the Intro.*

[39] *Does this mean bacterial symbionts of plants?*

[40] *Hard to know how to interpret this since interactions can involved multiple kingdoms.*

[41] *Important point, and I think you should bring in more of the literature documenting that E+ advantage increases in drier in environments, since our results show the opposite – we will then need to interpret why we get the opposite.*

[42] *I like all the ideas you have here, just think we need to address more directly how our results fit with or contradict other results in the literature, including grass-endo systems that show E+ advantage in drought.*
It is also striking to me that the Discussion never returns to the predictions you establish in the conceptual figure. If you want to keep the conceptual figure then the Discussion needs to revisit it.

References

Afkhami, M. E., McIntyre, P. J., and Strauss, S. Y. (2014). Mutualist-mediated effects on species' range limits across large geographic scales. *Ecology letters*, 17(10):1265–1273.

Afkhami, M. E. and Rudgers, J. A. (2008). Symbiosis lost: imperfect vertical transmission of fungal endophytes in grasses. *The American Naturalist*, 172(3):405–416.

Ahlholm, J. U., Helander, M., Lehtimäki, S., Wäli, P., and Saikkonen, K. (2002). Vertically transmitted fungal endophytes: different responses of host-parasite systems to environmental conditions. *Oikos*, 99(1):173–183.

Amijee, F., Tinker, P., and Stribley, D. (1989). The development of endomycorrhizal root systems: Vii. a detailed study of effects of soil phosphorus on colonization. *New Phytologist*, 111(3):435–446.

Atala, C., Acuña-Rodríguez, I. S., Torres-Díaz, C., and Molina-Montenegro, M. A. (2022). Fungal endophytes improve the performance of host plants but do not eliminate the growth/defence trade-off. *New Phytologist*, 235(2).

Bastias, D. A., Martínez-Ghersa, M. A., Ballaré, C. L., and Gundel, P. E. (2017). Epichloë fungal endophytes and plant defenses: not just alkaloids. *Trends in Plant Science*, 22(11):939–948.

Bertness, M. D. and Callaway, R. (1994). Positive interactions in communities. *Trends in ecology & evolution*, 9(5):191–193.

Bronstein, J. L. (1994). Conditional outcomes in mutualistic interactions. *Trends in ecology & evolution*, 9(6):214–217.

Bronstein, J. L. (2001). The exploitation of mutualisms. *Ecology letters*, 4(3):277–287.

Clay, K., Holah, J., and Rudgers, J. A. (2005). Herbivores cause a rapid increase in hereditary symbiosis and alter plant community composition. *Proceedings of the National Academy of Sciences*, 102(35):12465–12470.

Clay, K. and Schardl, C. (2002). Evolutionary origins and ecological consequences of endophyte symbiosis with grasses. *the american naturalist*, 160(S4):S99–S127.

Decunta, F. A., Pérez, L. I., Malinowski, D. P., Molina-Montenegro, M. A., and Gundel, P. E. (2021). A systematic review on the effects of epichloë fungal endophytes on drought tolerance in cool-season grasses. *Frontiers in Plant Science*, 12:644731.

Emery, S. M., Thompson, D., and Rudgers, J. A. (2010). Variation in endophyte symbiosis, herbivory and drought tolerance of ammophila breviligulata populations in the great lakes region. *The American Midland Naturalist*, 163(1):186–196.

Fowler, J. C., Donald, M. L., Bronstein, J. L., and Miller, T. E. (2023). The geographic footprint of mutualism: How mutualists influence species' range limits. *Ecological Monographs*, 93(1):e1558.

Fowler, J. C., Moutouama, J., and Miller, T. E. (2025). Increasing prevalence of plant-fungal symbiosis across two centuries of environmental change. *Global Change Biology*, 31(11):e70577.

GBIF (2025a). Occurrence download for *Agrostis hyemalis*. Accessed 5 November 2025.

GBIF (2025b). Occurrence download for *Elymus virginicus*. Accessed 5 November 2025.

GBIF (2025c). Occurrence download for *Poa autumnalis*. Accessed 5 November 2025.

Giauque, H., Connor, E. W., and Hawkes, C. V. (2019). Endophyte traits relevant to stress tolerance, resource use and habitat of origin predict effects on host plants. *New Phytologist*, 221(4):2239–2249.

Gundel, P. E., Rudgers, J. A., and Ghersa, C. M. (2011). Incorporating the process of vertical transmission into understanding of host–symbiont dynamics. *Oikos*, 120(8):1121–1128.

Gundel, P. E., Sun, P., Charlton, N. D., Young, C. A., Miller, T. E., and Rudgers, J. A. (2020). Simulated folivory increases vertical transmission of fungal endophytes that deter herbivores and alter tolerance to herbivory in poa autumnalis. *Annals of Botany*, 125(6):981–991.

Gundel, P. E., Ueno, A. C., Casas, C., Miller, T. E., Pérez, L. I., Cuyeu, R., and Omacini, M. (2024). Temporal host-symbiont dynamics in community contexts: Impacts of host fitness and vertical transmission efficiency on symbiosis prevalence. *Functional Ecology*, 38(12):2610–2622.

Hart, E. M. and Bell, K. (2015). *prism: Download data from the Oregon prism project*. R package version 0.0.6.

Hoang, K. L., Salguero-Gómez, R., Pike, V. L., and King, K. C. (2024). The impacts of host association and perturbation on symbiont fitness. *Symbiosis*, 92(3):439–451.

Hoeksema, J. D. and Bruna, E. M. (2015). Context-dependent outcomes of mutualistic interactions. *Mutualism*, 10:181–202.

Holmgren, M. and Scheffer, M. (2010). Strong facilitation in mild environments: the stress gradient hypothesis revisited. *Journal of Ecology*, 98(6):1269–1275.

Johnson, N. C., Graham, J. H., and Smith, F. A. (1997). Functioning of mycorrhizal associations along the mutualism–parasitism continuum. *The New Phytologist*, 135(4):575–585.

Klironomos, J. N. (2003). Variation in plant response to native and exotic arbuscular mycorrhizal fungi. *Ecology*, 84(9):2292–2301.

Leuchtmann, A., Bacon, C. W., Schardl, C. L., White Jr, J. F., and Tadych, M. (2014). Nomenclatural realignment of neotyphodium species with genus epichloë. *Mycologia*, 106(2):202–215.

Liu, M. (2023). Fungi canadenses no. 351: Epichloe elymi. *Canadian Journal of Plant Pathology*, 45(3):300–303.

Louthan, A. M., Doak, D. F., and Angert, A. L. (2015). Where and when do species interactions set range limits? *Trends in ecology & evolution*, 30(12):780–792.

Oberhofer, M., Güsewell, S., and Leuchtmann, A. (2014). Effects of natural hybrid and non-hybrid epichloë endophytes on the response of hordelymus europaeus to drought stress. *New Phytologist*, 201(1):242–253.

Porter, S. S., Bantay, R., Friel, C. A., Garoutte, A., Gdanetz, K., Ibarreta, K., Moore, B. M., Shetty, P., Siler, E., and Friesen, M. L. (2020). Beneficial microbes ameliorate abiotic and biotic sources of stress on plants. *Functional Ecology*, 34(10):2075–2086.

R Core Team (2023). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Rudgers, J. A., Afkhami, M. E., Bell-Dereske, L., Chung, Y. A., Crawford, K. M., Kivlin, S. N., Mann, M. A., and Nuñez, M. A. (2020). Climate disruption of plant-microbe interactions. *Annual review of ecology, evolution, and systematics*, 51:561–586.

Rudgers, J. A., Fletcher, R. A., Olivas, E., Young, C. A., Charlton, N. D., Pearson, D. E., and Maron, J. L. (2016). Long-term ungulate exclusion reduces fungal symbiont prevalence in native grasslands. *Oecologia*, 181(4):1151–1161.

Rudgers, J. A., Miller, T. E., Ziegler, S. M., and Craven, K. D. (2012). There are many ways to be a mutualist: endophytic fungus reduces plant survival but increases population growth. *Ecology*, 93(3):565–574.

Ruiz-lazona, J. M., Azcón, R., and Palma, J. (1996). Superoxide dismutase activity in arbuscular mycorrhizal lactuca sativa plants subjected to drought stress. *New Phytologist*, 134(2):327–333.

Saikkonen, K., Faeth, S. H., Helander, M., and Sullivan, T. (1998). Fungal endophytes: a continuum of interactions with host plants. *Annual review of Ecology and Systematics*, 29(1):319–343.

Saikkonen, K., Saari, S., and Helander, M. (2010). Defensive mutualism between plants and endophytic fungi? *Fungal Diversity*, 41(1):101–113.

Saikkonen, K., Wäli, P., Helander, M., and Faeth, S. H. (2004). Evolution of endophyte–plant symbioses. *Trends in plant science*, 9(6):275–280.

Schardl, C. L., Leuchtmann, A., and Spiering, M. J. (2004). Symbioses of grasses with seedborne fungal endophytes. *Annu. Rev. Plant Biol.*, 55(1):315–340.

Shaw, R. B. (2011). *Guide to Texas grasses*. Texas A&M University Press.

Silva, L. A. and Zanella, G. (2024). Robust leave-one-out cross-validation for high-dimensional bayesian models. *Journal of the American Statistical Association*, 119(547):2369–2381.

Soudzilovskaia, N. A., van der Heijden, M. G., Cornelissen, J. H., Makarov, M. I., Onipchenko, V. G., Maslov, M. N., Akhmetzhanova, A. A., and van Bodegom, P. M. (2015). Quantitative assessment of the differential impacts of arbuscular and ectomycorrhiza on soil carbon cycling. *New Phytologist*, 208(1):280–293.

Stan Development Team (2024). RStan: the R interface to Stan. R package version 2.32.6.

Thirkell, T. J., Pastok, D., and Field, K. J. (2020). Carbon for nutrient exchange between arbuscular mycorrhizal fungi and wheat varies according to cultivar and changes in atmospheric carbon dioxide concentration. *Global change biology*, 26(3):1725–1738.

Vega, F. E. (2008). Insect pathology and fungal endophytes. *Journal of invertebrate pathology*, 98(3):277–279.

Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and computing*, 27:1413–1432.

White Jr, J. F. (1994). Endophyte-host associations in grasses. xx. structural and reproductive studies of epichloë amarillans sp. nov. and comparisons to e. typhina. *Mycologia*, 86(4):571–580.

Yule, K. M., Miller, T. E., and Rudgers, J. A. (2013). Costs, benefits, and loss of vertically transmitted symbionts affect host population dynamics. *Oikos*, 122(10):1512–1520.

Figures

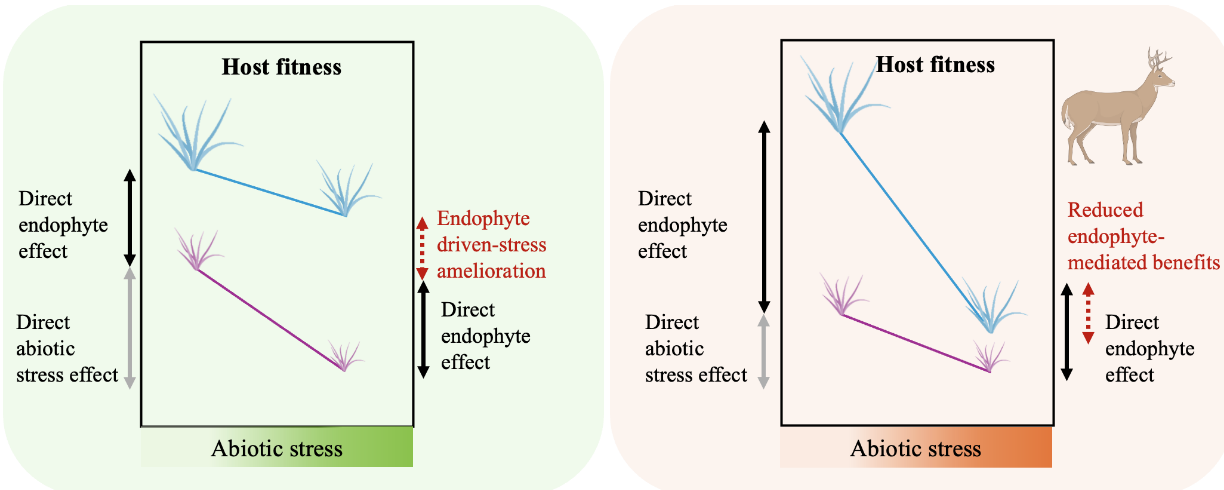

Figure 1: Conceptual framework illustrating the predicted effects of abiotic stress, biotic stress, and endophyte-mediated mutualism on host plant fitness (adapted from (Porter et al., 2020)). (a) **Endophyte-driven stress amelioration scenario:** Endophytes are predicted to enhance host performance under increasing abiotic stress (e.g., low precipitation) in the absence of herbivores by mitigating drought-related physiological costs. (b) **Reduced endophyte-mediated benefits scenario:** When abiotic stress is combined with herbivory, the positive effect of endophytes is expected to weaken. Limited plant resources are divided between defense against herbivores and sustaining the endophyte, reducing the endophyte's capacity to ameliorate drought stress and potentially resulting in a net cost of hosting an endophyte. Colors are used solely for visualization: blue lines represent endophyte-infected plants and violet lines represent endophyte-free plants.

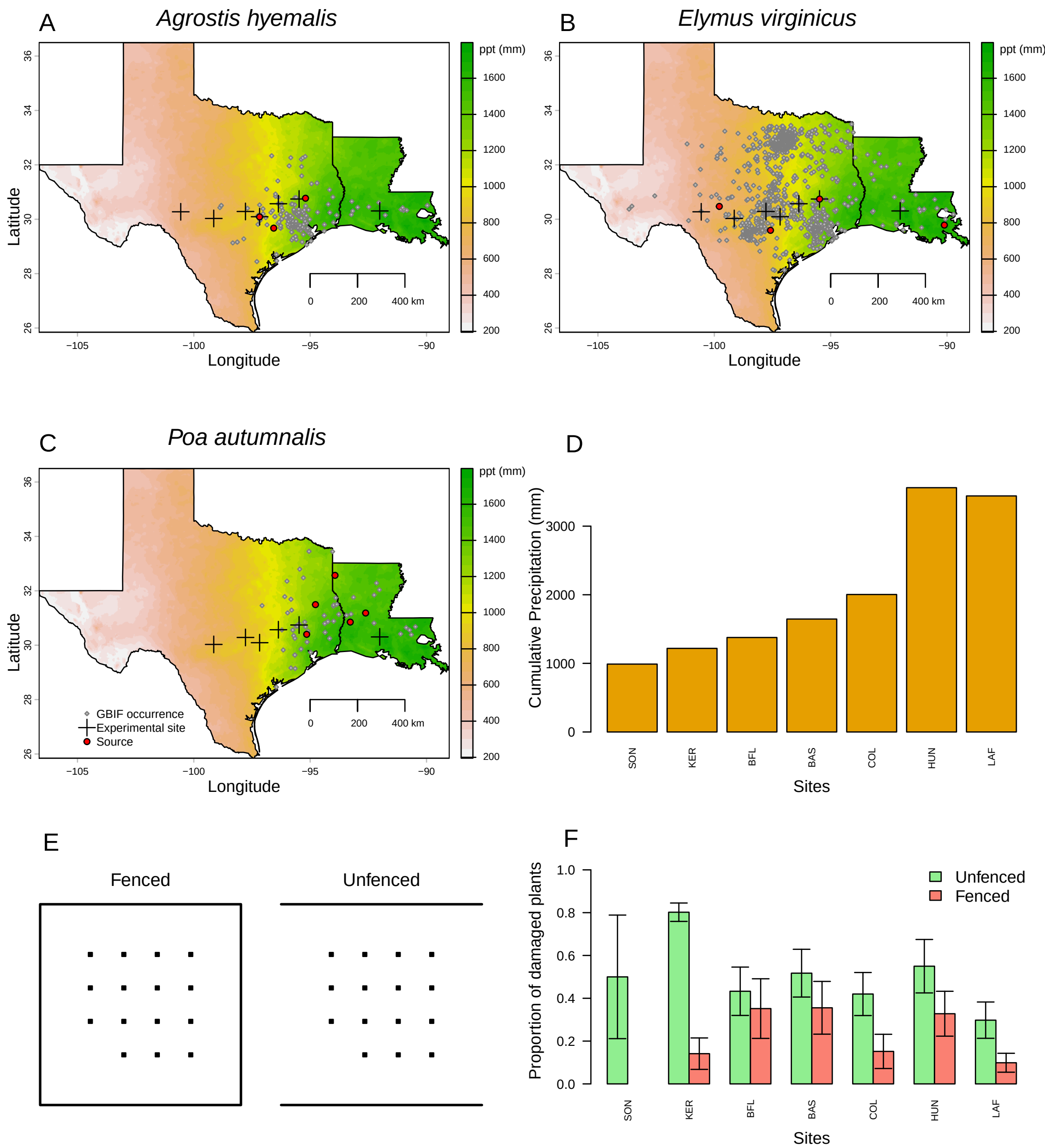

Figure 2: Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals of climate predictions (1996–2025) from PRISM data (Hart and Bell, 2015), illustrating the precipitation gradient across experimental sites (scale on the right is in mm). Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). (D) Annual cumulative precipitation during each census year (May 2023 – June 2025) at each experimental site, ordered west to east. Values are also derived from PRISM climate data. (E) Schematic illustration of fenced (left) and unfenced (right) plots used to assess herbivory. Black squares represent individual plants; plot shape correspond to experimental design. (F) Proportion of damaged plants in fenced (salmon) and unfenced (light green) plots across experimental sites, with mean ± standard error. Sites are ordered west to east as in panel D. Panels E–F illustrate herbivory treatments and outcomes, and species-specific breakdowns are provided in Supplementary Figures.

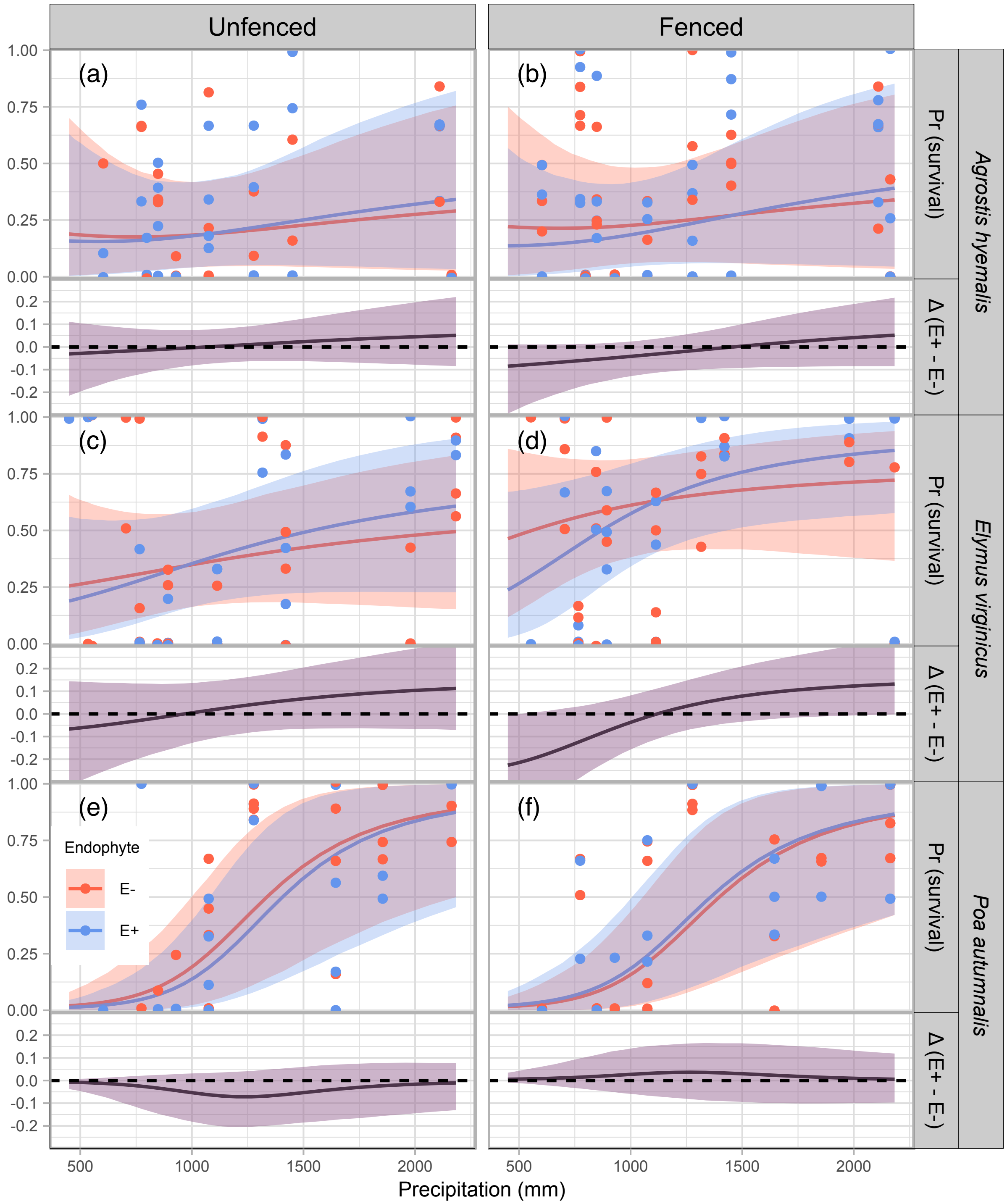

Figure 3: Predicted survival rate (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

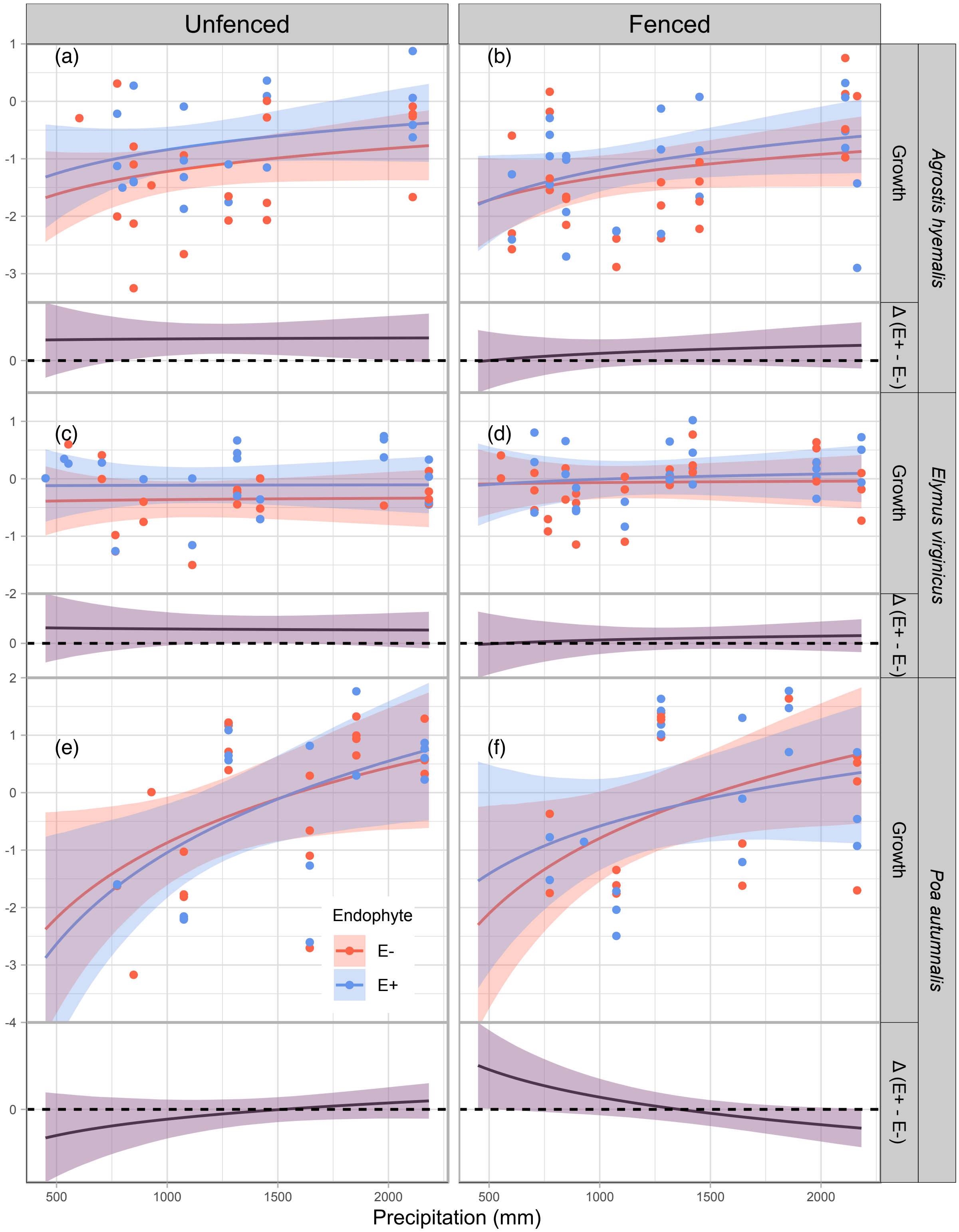

Figure 4: Predicted relative growth from year to the next year (top panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed growth per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

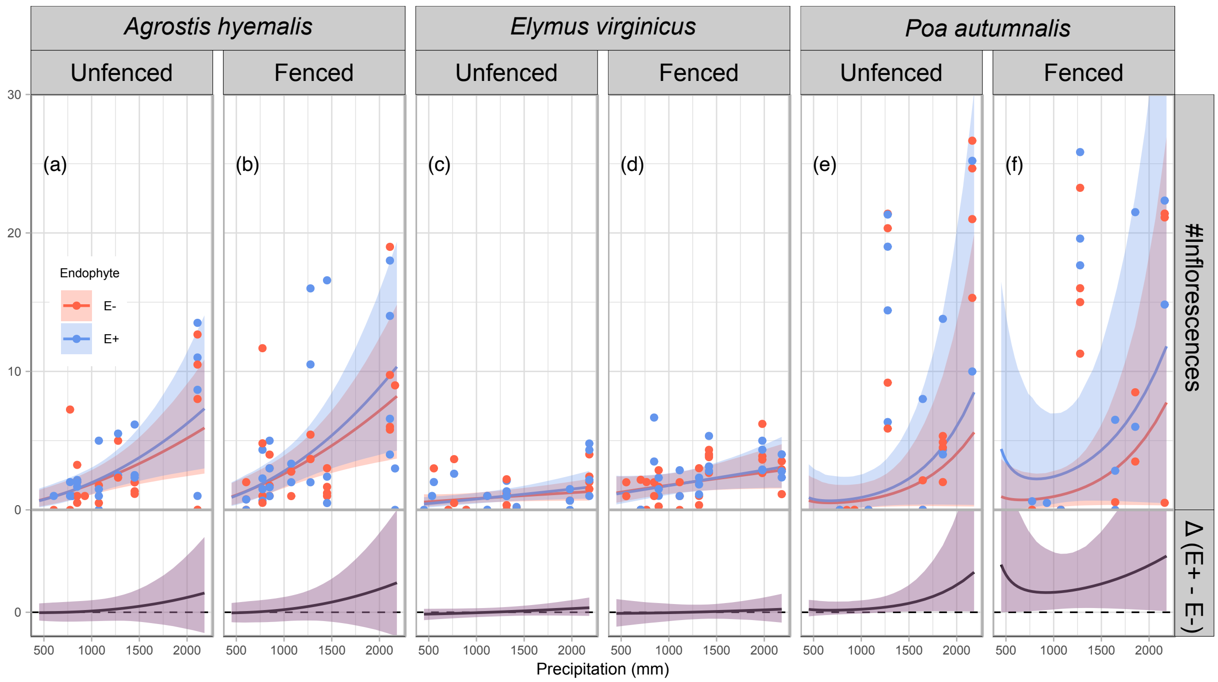

Figure 5: Predicted number of inflorescences (upper panels) and the difference between endophyte-infected (E^+) and endophyte-free (E^-) plants ($\Delta(E^+ – E^-)$, lower panels) are shown across a gradient of precipitation. Shaded areas represent 90% credible intervals. Points indicate observed means, jittered for visualization. Panels are grouped by species and herbivory treatment (Fenced, Unfenced).

Tables

Table 1: Model comparison of drivers of grass- endophyte symbiosis across four fitness components (vital rates). ΔELPD indicates the difference in expected log predictive density relative to the best-fitting model; SE is the standard error of the difference.

Vital rate	**Model type**	**ΔELPD**	**SE**
Survival	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.11	2.81
	Endophyte × Climate × Herbivory	−4.98	1.89
Growth	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.19	2.81
	Endophyte × Climate × Herbivory	−1.81	2.75
Inflorescence	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.62	3.53
	Endophyte × Climate × Herbivory	−3.78	2.63
Spikelet	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.01	3.82
	Endophyte × Climate × Herbivory	−3.16	2.43

S.1 Supporting Information

S.1.1 Supporting Methods

Candidate models for abiotic and biotic drivers of cool-season grass species

1. **Endophyte × Climate model:** includes endophyte status (endo), precipitation (P), and their interaction:

$$\begin{aligned}\mu_i = &\beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i]endo_i + \beta_P[\mathrm{Sp}_i]P_i + \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i]endo_iP_i \\ &+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}\end{aligned}$$

2. **Endophyte × Herbivory model:** includes endophyte status (endo), herbivory (herb), and their interaction:

$$\begin{aligned}\mu_i = &\beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i]endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i]herb_i + \beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i]endo_iherb_i \\ &+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}\end{aligned}$$

3. **Endophyte × Climate × Herbivory:** combines all predictors and interactions up to the three-way interaction:

$$\begin{aligned}\mu_i = &\beta_0[\mathrm{Sp}_i] + \beta_P[\mathrm{Sp}_i]P_i + \beta_{\mathrm{endo}}[\mathrm{Sp}_i]endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i]herb_i \\ &+ \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i]endo_iP_i + \beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i]herb_iP_i + \beta_{\mathrm{herb}\times\mathrm{endo}}[\mathrm{Sp}_i]herb_iendo_i \\ &+ \beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i]endo_iherb_iP_i + \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}\end{aligned}$$

In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\mathrm{Sp}_i]$ corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors. $\beta_{\mathrm{endo}}[\mathrm{Sp}_i]$ and $\beta_{\mathrm{herb}}[\mathrm{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively. The linear effect of precipitation is captured by $\beta_P[\mathrm{Sp}_i]$, while interactions are represented by $\beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i]$, and the three-way interaction $\beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i]$. All models include hierarchical random effects for site-year, source population, and plot, with species-specific random effects $\phi_{\mathrm{site}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{site}})$, $\omega_{\mathrm{source}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{source}})$, and $\rho_{\mathrm{plot}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{plot}})$.

S.1.2 Supporting Tables

Table S1: Geographic coordinates and brief descriptions of source populations for *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

Species	Population	Latitude	Longitude	Description
AGHY	SJC	30.771969	-95.200620	San Jacinto County, TX
AGHY	COHR	29.672833	-96.567517	Columbus, TX
AGHY	LPEF	30.085383	-97.172508	Lost Pines
POAU	LA7	30.849660	-93.276772	DeRidder, LA
POAU	LA1	32.568125	-93.932976	Jean Lafitte National Historical Park and Preserve[43]
POAU	LA2	31.183040	-92.616969	Jean Lafitte National Historical Park and Preserve
POAU	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX
POAU	WB	30.399346	-95.150026	Sam Houston National Forest
ELVI	HUNT	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	SHS[44]	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	RL5	30.471717	-99.783803	Texas Tech University Field Station, Junction
ELVI	PALM	29.591623	-97.584781	Palmetto State Park, TX
ELVI	JLP	29.785105	-90.114933	Jean Lafitte Park, outside of New Orleans

Table S2: Common garden sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
1335 Medina Hwy, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S3: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0	0.000
Survival	Linear	-1.230	0.947
Growth	Quadratic	0	0.000
Growth	Linear	-0.544	0.924
Flowering	Quadratic	0	0.000
Flowering	Linear	-0.763	0.956
Spikelet	Quadratic	0	0.000
Spikelet	Linear	-0.097	0.957

Table S4: Posterior summaries of Δ ($E^+ - E^-$) for survival at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
Agrostis hyemalis	Unfenced	765.76	-0.01055	-0.12081	0.07964	0.3723
Agrostis hyemalis	Unfenced	848.28	-0.00814	-0.10368	0.07597	0.3980
Agrostis hyemalis	Unfenced	1113.25	0.00173	-0.07149	0.07904	0.5239
Agrostis hyemalis	Unfenced	1644.36	0.02230	-0.06632	0.14693	0.6981
Agrostis hyemalis	Unfenced	2163.28	0.03508	-0.08410	0.21801	0.7237
Agrostis hyemalis	Fenced	765.76	-0.04897	-0.17775	0.01302	0.0961
Agrostis hyemalis	Fenced	848.28	-0.04477	-0.15803	0.01516	0.1076
Agrostis hyemalis	Fenced	1113.25	-0.02642	-0.11372	0.03630	0.2266
Agrostis hyemalis	Fenced	1644.36	0.01141	-0.08847	0.12851	0.5950
Agrostis hyemalis	Fenced	2163.28	0.03789	-0.08484	0.21422	0.7222
Elymus virginicus	Unfenced	765.76	-0.02574	-0.19555	0.13388	0.3799
Elymus virginicus	Unfenced	848.28	-0.01565	-0.17000	0.13230	0.4250
Elymus virginicus	Unfenced	1113.25	0.01937	-0.10368	0.14589	0.6070
Elymus virginicus	Unfenced	1644.36	0.07749	-0.06409	0.22970	0.8217
Elymus virginicus	Unfenced	2163.28	0.10622	-0.07003	0.31076	0.8445
Elymus virginicus	Fenced	765.76	-0.12411	-0.28674	0.03254	0.0992
Elymus virginicus	Fenced	848.28	-0.09198	-0.23593	0.04709	0.1418
Elymus virginicus	Fenced	1113.25	-0.00179	-0.11213	0.11041	0.4882
Elymus virginicus	Fenced	1644.36	0.08855	-0.02165	0.24020	0.9069
Elymus virginicus	Fenced	2163.28	0.11327	-0.00261	0.32611	0.9456
Poa autumnalis	Unfenced	765.76	-0.00979	-0.12238	0.02097	0.1853
Poa autumnalis	Unfenced	848.28	-0.01778	-0.14683	0.02450	0.1735
Poa autumnalis	Unfenced	1113.25	-0.05636	-0.20042	0.03503	0.1429
Poa autumnalis	Unfenced	1644.36	-0.02891	-0.17290	0.07018	0.2804
Poa autumnalis	Unfenced	2163.28	-0.00174	-0.13201	0.07681	0.4319
Poa autumnalis	Fenced	765.76	0.00399	-0.03466	0.09583	0.6685
Poa autumnalis	Fenced	848.28	0.00739	-0.04282	0.11317	0.6803
Poa autumnalis	Fenced	1113.25	0.02557	-0.06906	0.15496	0.7063
Poa autumnalis	Fenced	1644.36	0.01678	-0.09915	0.15764	0.6297
Poa autumnalis	Fenced	2163.28	0.00181	-0.09727	0.12122	0.5663

Table S5: Posterior summaries of Δ ($E^{+} - E^{-}$) for growth across precipitation quantiles for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes; negative medians indicate costly effects.

Species	**Herbivore exclusion**	**Precipitation (mm)**	**Median Δ**	**Lower 90%**	**Upper 90%**	**P($\Delta > 0$)**
Agrostis hyemalis	Unfenced	773.91	0.37006	-0.00924	0.75437	0.9456
Agrostis hyemalis	Unfenced	1075.09	0.37940	0.10078	0.65111	0.9866
Agrostis hyemalis	Unfenced	1316.25	0.38541	0.11668	0.64408	0.9915
Agrostis hyemalis	Unfenced	1979.35	0.39167	0.01393	0.77022	0.9539
Agrostis hyemalis	Unfenced	2163.28	0.39436	-0.02748	0.81310	0.9400
Agrostis hyemalis	Fenced	773.91	0.07955	-0.23563	0.39683	0.6612
Agrostis hyemalis	Fenced	1075.09	0.13909	-0.10277	0.37262	0.8279
Agrostis hyemalis	Fenced	1316.25	0.17455	-0.07319	0.41710	0.8781
Agrostis hyemalis	Fenced	1979.35	0.24676	-0.11528	0.60922	0.8733
Agrostis hyemalis	Fenced	2163.28	0.26153	-0.13527	0.66118	0.8657
Elymus virginicus	Unfenced	773.91	0.25869	-0.11938	0.62862	0.8693
Elymus virginicus	Unfenced	1075.09	0.24977	-0.02146	0.51978	0.9343
Elymus virginicus	Unfenced	1316.25	0.24421	0.00389	0.48413	0.9530
Elymus virginicus	Unfenced	1979.35	0.23379	-0.05653	0.52131	0.9096
Elymus virginicus	Unfenced	2163.28	0.23175	-0.08415	0.54666	0.8877
Elymus virginicus	Fenced	773.91	0.03164	-0.30420	0.37363	0.5644
Elymus virginicus	Fenced	1075.09	0.06458	-0.16044	0.29000	0.6825
Elymus virginicus	Fenced	1316.25	0.08512	-0.10777	0.27375	0.7682
Elymus virginicus	Fenced	1979.35	0.12472	-0.12987	0.37637	0.7903
Elymus virginicus	Fenced	2163.28	0.13352	-0.15046	0.41561	0.7788
Poa autumnalis	Unfenced	773.91	-0.27505	-0.74845	0.20422	0.1688
Poa autumnalis	Unfenced	1075.09	-0.14041	-0.44570	0.16567	0.2237
Poa autumnalis	Unfenced	1316.25	-0.05910	-0.29128	0.17786	0.3413
Poa autumnalis	Unfenced	1979.35	0.10761	-0.15864	0.37505	0.7388
Poa autumnalis	Unfenced	2163.28	0.14420	-0.16050	0.44856	0.7749
Poa autumnalis	Fenced	773.91	0.39049	-0.05133	0.83526	0.9270
Poa autumnalis	Fenced	1075.09	0.16142	-0.12313	0.45470	0.8212
Poa autumnalis	Fenced	1316.25	0.02003	-0.20891	0.25959	0.5587
Poa autumnalis	Fenced	1979.35	-0.26222	-0.54759	0.03589	0.0723
Poa autumnalis	Fenced	2163.28	-0.32245	-0.64896	0.01200	0.0558

Table S6: Posterior summaries of Δ ($E^{+} - E^{-}$) for flowering output at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
Agrostis hyemalis	Unfenced	773.91	-0.6824	-0.6904	0.7822	0.4889
Agrostis hyemalis	Unfenced	1075.09	0.0286	-0.6483	0.9444	0.6018
Agrostis hyemalis	Unfenced	1316.25	0.4663	-0.6218	1.2633	0.6909
Agrostis hyemalis	Unfenced	1979.35	1.3488	-1.2041	4.0378	0.7552
Agrostis hyemalis	Unfenced	2163.28	1.5409	-1.4924	5.3973	0.7522
Agrostis hyemalis	Fenced	773.91	-0.6824	-0.7351	0.9077	0.5238
Agrostis hyemalis	Fenced	1075.09	0.0286	-0.6302	1.2286	0.6696
Agrostis hyemalis	Fenced	1316.25	0.4663	-0.6108	1.7763	0.7609
Agrostis hyemalis	Fenced	1979.35	1.3488	-1.3208	5.6835	0.7965
Agrostis hyemalis	Fenced	2163.28	1.5409	-1.7235	7.3637	0.7922
Elymus virginicus	Unfenced	773.91	-0.6824	-0.4386	0.2723	0.3289
Elymus virginicus	Unfenced	1075.09	0.0286	-0.3390	0.3120	0.4611
Elymus virginicus	Unfenced	1316.25	0.4663	-0.2792	0.3903	0.5915
Elymus virginicus	Unfenced	1979.35	1.3488	-0.2491	0.8798	0.7902
Elymus virginicus	Unfenced	2163.28	1.5409	-0.2647	1.0691	0.8082
Elymus virginicus	Fenced	773.91	-0.6824	-0.8003	0.7325	0.4283
Elymus virginicus	Fenced	1075.09	0.0286	-0.6057	0.5997	0.4688
Elymus virginicus	Fenced	1316.25	0.4663	-0.4972	0.5628	0.5208
Elymus virginicus	Fenced	1979.35	1.3488	-0.6334	1.0660	0.6272
Elymus virginicus	Fenced	2163.28	1.5409	-0.7473	1.3341	0.6318
Poa autumnalis	Unfenced	773.91	-0.6824	-0.1587	0.7919	0.7062
Poa autumnalis	Unfenced	1075.09	0.0286	-0.0711	0.9392	0.8638
Poa autumnalis	Unfenced	1316.25	0.4663	0.0137	1.3447	0.9629
Poa autumnalis	Unfenced	1979.35	1.3488	0.0793	6.8120	0.9926
Poa autumnalis	Unfenced	2163.28	1.5409	0.0645	10.6951	0.9858
Poa autumnalis	Fenced	773.91	-0.6824	0.0907	5.6618	0.9981
Poa autumnalis	Fenced	1075.09	0.0286	0.2492	4.3096	0.9998
Poa autumnalis	Fenced	1316.25	0.4663	0.3164	4.7790	0.9999
Poa autumnalis	Fenced	1979.35	1.3488	0.1612	11.8419	0.9976
Poa autumnalis	Fenced	2163.28	1.5409	0.0888	15.3801	0.9848

Table S7: Posterior summaries of Δ ($E^{+} - E^{-}$) at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
Elymus virginicus	Unfenced	1113.25	0.1040	-4.4921	2.4788	0.3213
Elymus virginicus	Unfenced	1276.99	0.4008	-3.6975	2.7993	0.4012
Elymus virginicus	Unfenced	1644.36	0.9477	-2.5664	3.8884	0.6089
Elymus virginicus	Unfenced	2163.28	1.5409	-2.0652	6.1715	0.7733
Elymus virginicus	Unfenced	2182.71	1.5603	-2.0610	6.2851	0.7760
Elymus virginicus	Fenced	1113.25	0.1040	-4.2244	1.7020	0.2288
Elymus virginicus	Fenced	1276.99	0.4008	-3.0122	2.0977	0.3714
Elymus virginicus	Fenced	1644.36	0.9477	-1.2591	3.8036	0.7853
Elymus virginicus	Fenced	2163.28	1.5409	-0.4058	7.9946	0.9282
Elymus virginicus	Fenced	2182.71	1.5603	-0.3886	8.1667	0.9288
Poa autumnalis	Unfenced	1113.25	0.1040	-3.2874	1.7757	0.3134
Poa autumnalis	Unfenced	1276.99	0.4008	-2.7023	1.8008	0.3654
Poa autumnalis	Unfenced	1644.36	0.9477	-1.7415	2.3862	0.5832
Poa autumnalis	Unfenced	2163.28	1.5409	-1.9577	5.4441	0.7572
Poa autumnalis	Unfenced	2182.71	1.5603	-1.9811	5.5917	0.7592
Poa autumnalis	Fenced	1113.25	0.1040	0.3203	6.9989	0.9668
Poa autumnalis	Fenced	1276.99	0.4008	0.6863	6.6041	0.9828
Poa autumnalis	Fenced	1644.36	0.9477	0.8105	6.7216	0.9873
Poa autumnalis	Fenced	2163.28	1.5409	-0.9449	8.6686	0.8957
Poa autumnalis	Fenced	2182.71	1.5603	-1.0369	8.7690	0.8906

S.1.3 Supporting Figures

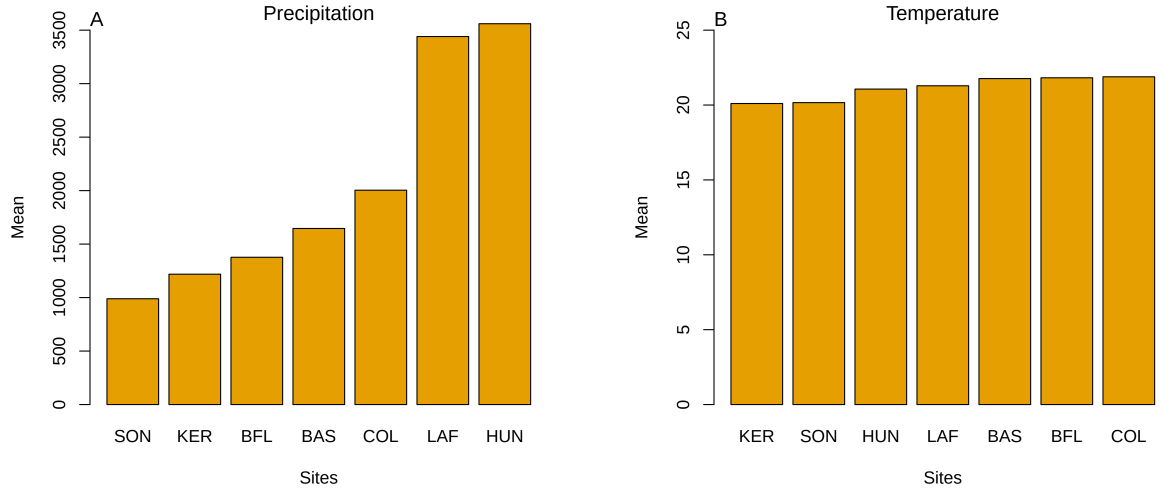

Fig S1: Abiotic conditions across the seven study sites. (A) Mean annual precipitation showing the pronounced east–west aridity gradient from mesic Louisiana to arid Texas. (B) Mean annual temperature across sites, which does not differ much. Panels highlight that precipitation, rather than temperature, constitutes the primary abiotic gradient in this system.

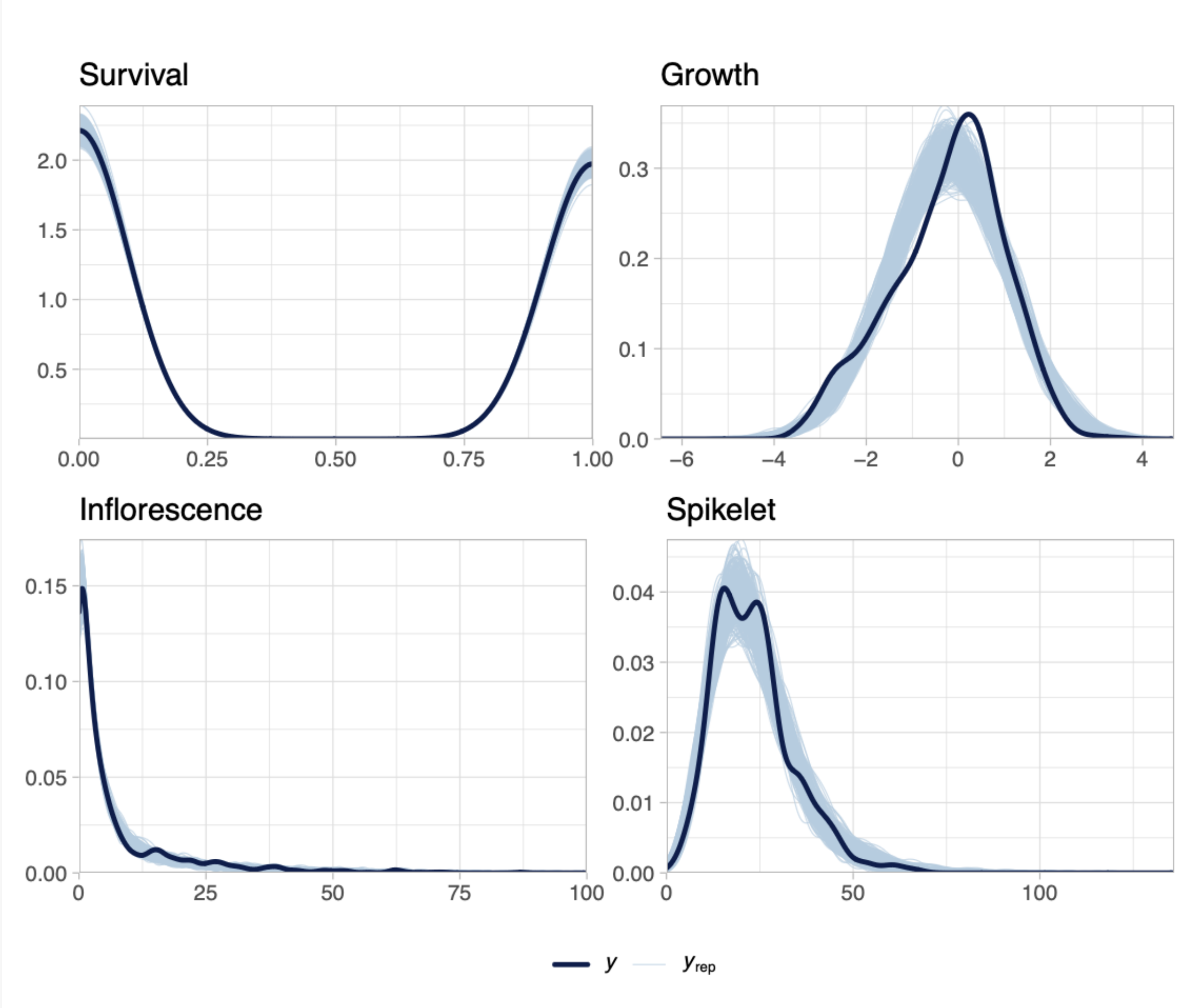

Fig S2: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

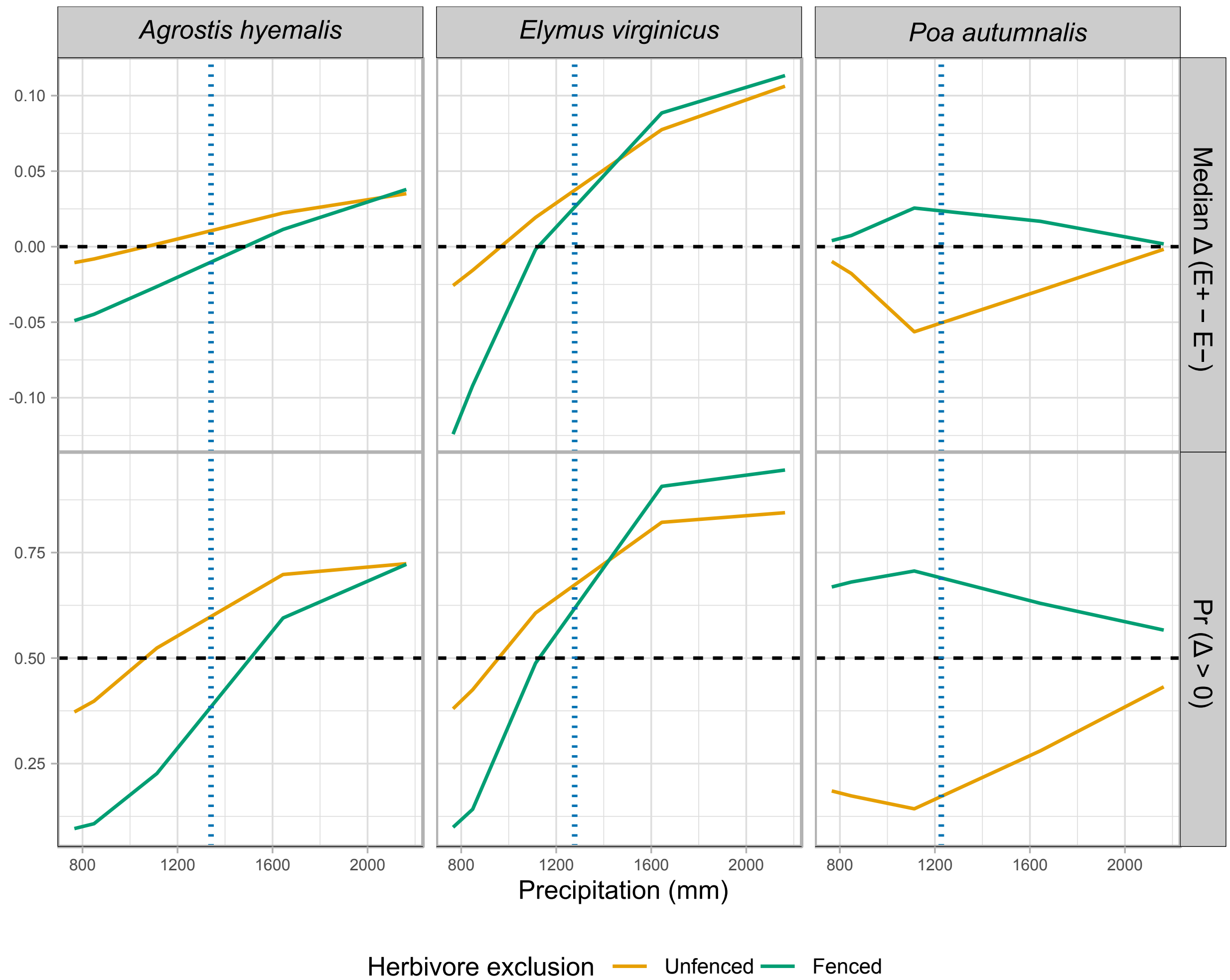

Fig S3: Predicted effects of endophyte presence (E^+) versus absence (E^-) on survival across precipitation gradients for three cool-season grasses. Solid lines show the median Δ ($E^+ - E^-$),or the posterior probability that $\Delta > 0$ under fenced and unfenced conditions. Horizontal dashed lines indicate no effect ($\Delta = 0$) or a 50% probability of benefit. Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

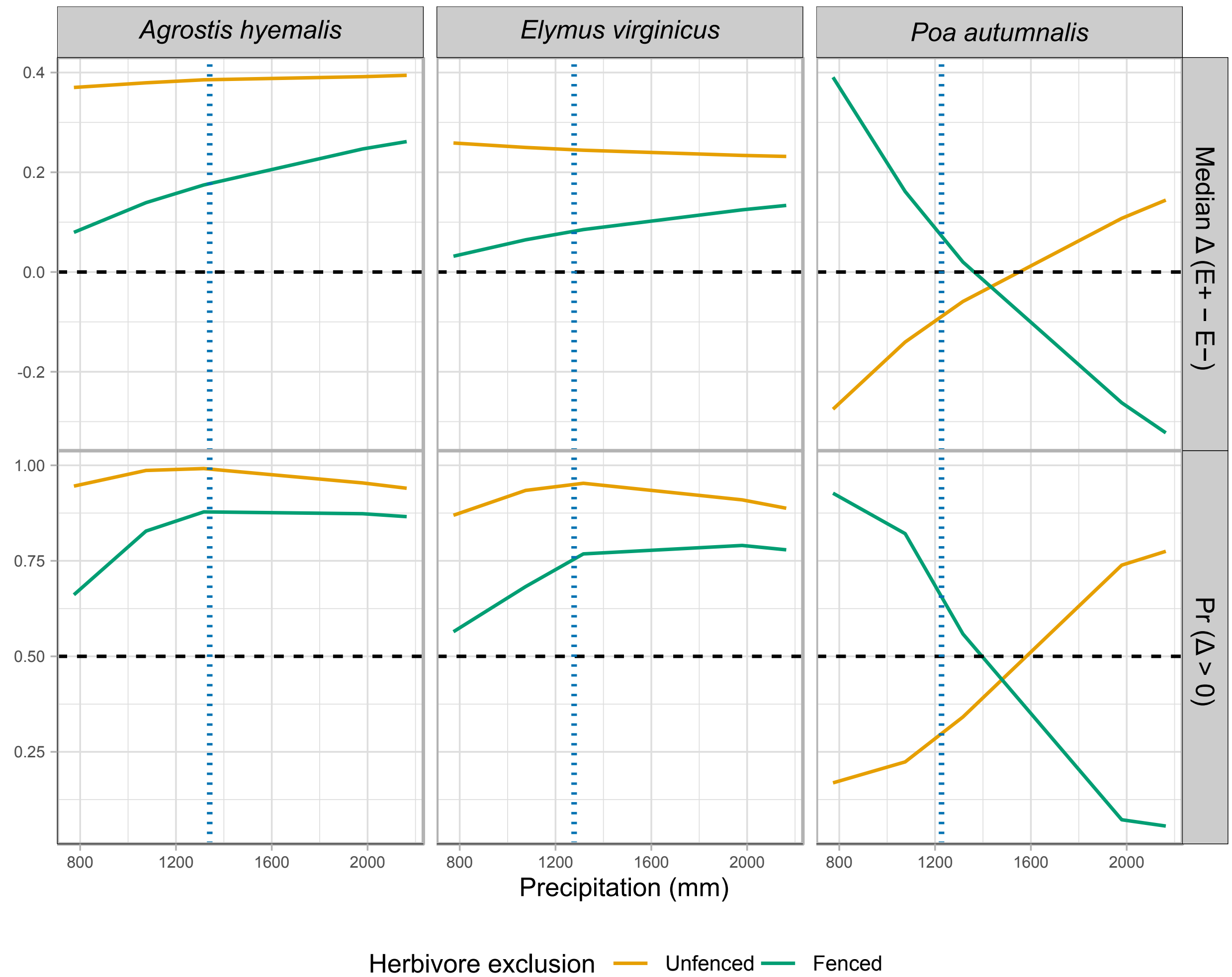

Fig S4: Effects of precipitation and endophyte presence on growth across three cool-season grass species. Lines show the predicted difference in growth between endophyte-present (E^+) and endophyte-absent (E^-) plants (Median Δ, top row) and the posterior probability that this difference is positive ($\Pr(\Delta > 0)$, bottom row) under fenced and unfenced treatments. Predictions were calculated at the 10th, 25th, 50th, 75th, and 90th percentiles of precipitation observed in the study. Horizontal dashed lines indicate $\Delta = 0$ (top row) and $\Pr(\Delta > 0)$ = 0.5 (bottom row). Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

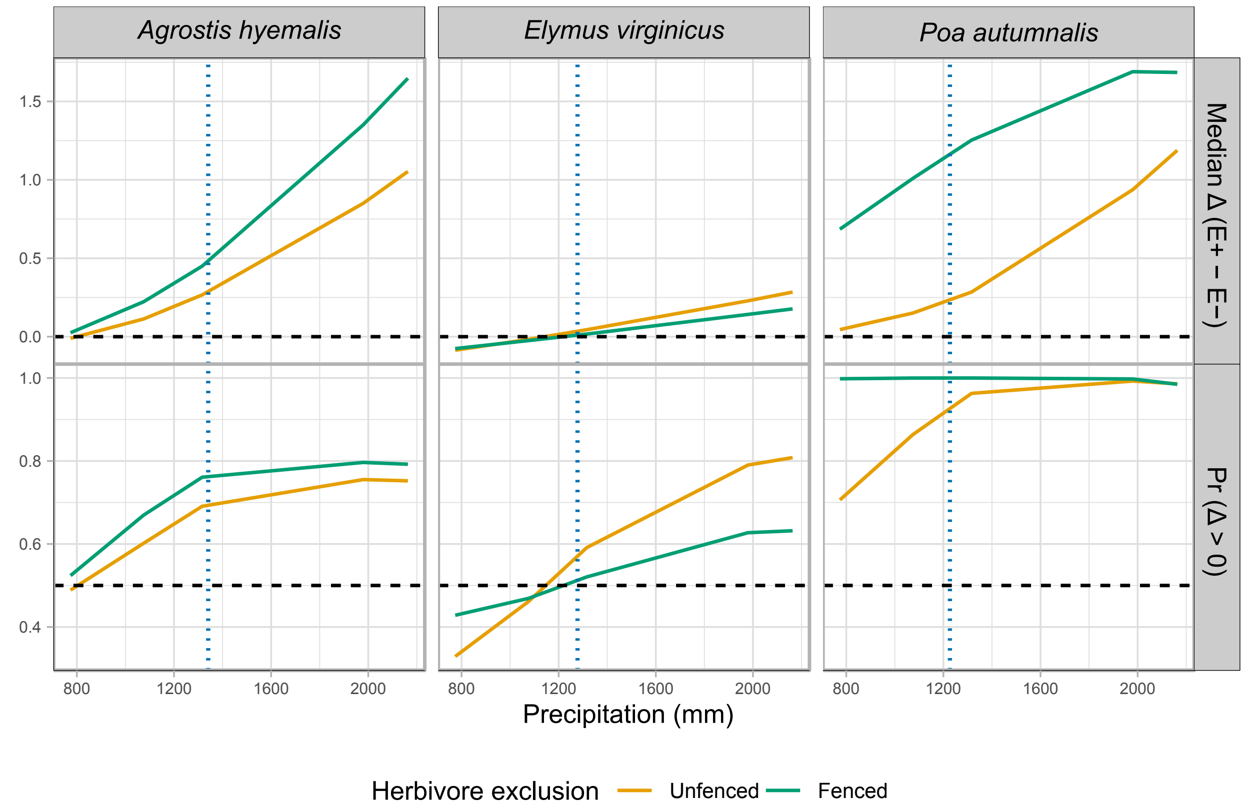

Fig S5: Effects of endophyte infection on flowering output across a precipitation gradient for three cool-season grass species. Lines show the median posterior difference Δ ($E^+ - E^-$) and the posterior probability that $\Delta > 0$ under fenced (herbivores excluded) and unfenced (herbivores present) conditions. Horizontal dashed lines indicate $\Delta = 0$ for median differences and $\Pr(\Delta > 0) = 0.5$. Endophyte effects were generally positive for *Agrostis hyemalis* and *Poa autumnalis*, but for *Elymus virginicus* benefits were context-dependent, with costs at lower precipitation

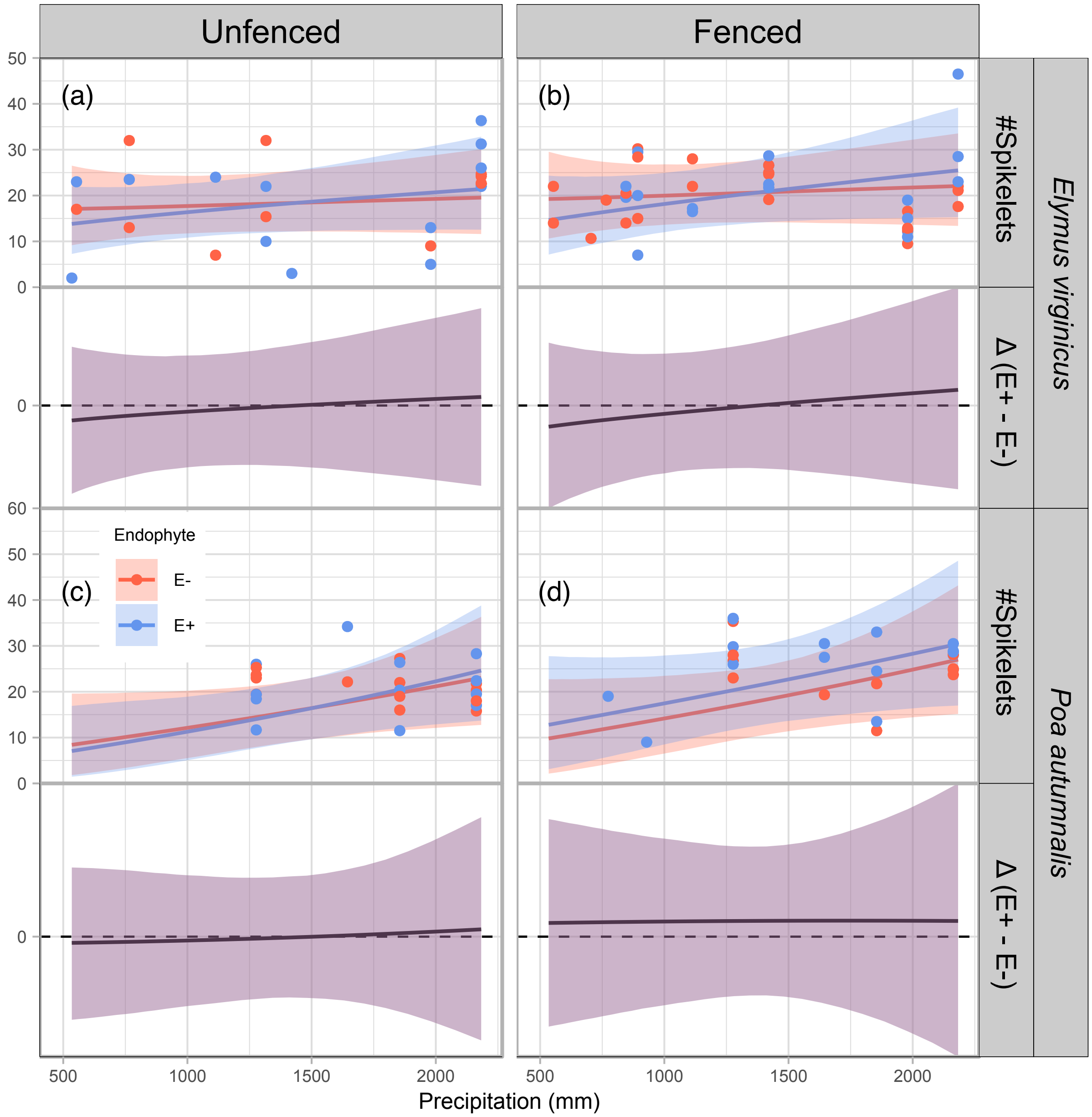

Fig S6: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

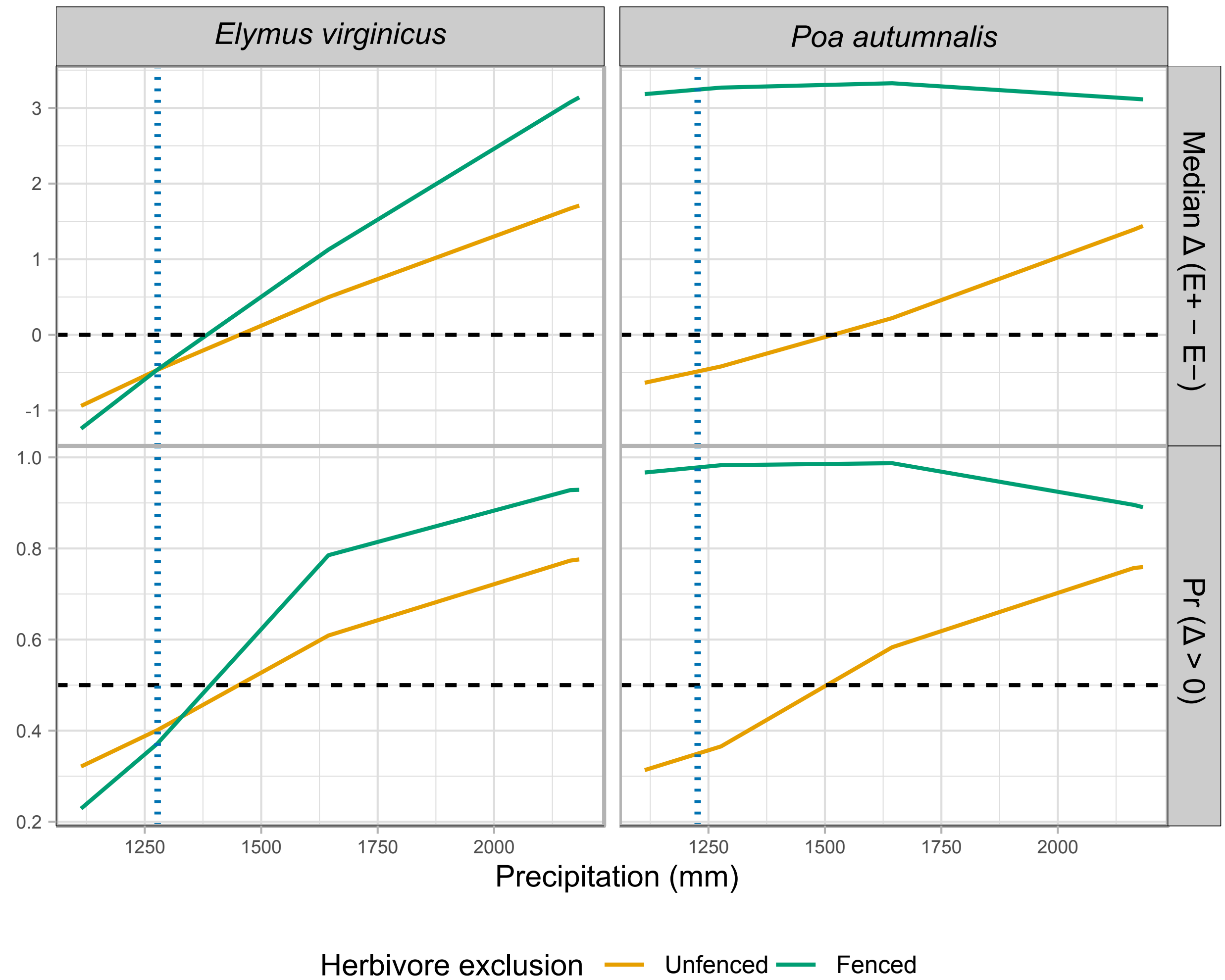

Fig S7: Effects of precipitation and endophyte presence on spikelet (reproductive output) for three cool-season grass species. Median Δ ($E^+ - E^-$) and posterior probability $Pr(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $Pr(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.